Team -8

Stage – 2-2: Clustering Analysis

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1. Introduction and Objectives:

The purpose of this clustering analysis is to identify different groups of customers based on their demographic characteristics, service subscriptions, customer tenure and monthly charges. The analysis aims to understand customer behaviour, identify patterns within the customer base and determine which customer segments are more likely to discontinue their services.

As part of this project, K-Means clustering was implemented using Python and Scikit-learn. The analysis builds on the data preparation process completed in Part 1 by the Data Engineer. The processed dataset was used as the foundation for developing customer segments.

From a business analyst's perspective, customer segmentation helps the organisation understand that not all customers have the same requirements or churn behaviour. By identifying distinct customer groups, the business can develop targeted retention strategies rather than applying the same approach to every customer.

The main objectives of this analysis are:

- Identify meaningful customer segments using unsupervised machine learning.
- Understand the characteristics of each customer segment.
- Examine the relationship between customer segments and churn rates.
- Identify high-risk customer groups that require retention strategies.
- Provide actionable business recommendations based on the clustering results.
- Prepare cluster information that can potentially support the subsequent predictive modelling stage.

2. CLUSTERING METHODOLOGY

The initial assessment confirmed that the dataset was suitable for clustering. There were no missing values or inconsistent categorical labels. Although duplicate records existed, they were retained because there was insufficient information to establish that they represented errors.

Additional Data Processing for Clustering

K-Means is a distance-based algorithm that requires numerical input. Therefore, some additional transformations were required before applying the clustering model.

The processed dataset already contained standardised numerical variables and one-hot encoded categorical variables. However, some columns were still stored as text or Boolean values.

To avoid modifying the original processed dataset, a separate copy was created using:

```
df = processed.copy()
```

The following transformations were then performed.

3. ADDITIONAL DATA PROCESSING FOR CLUSTERING

1. Converting Categorical variables into numerical values

The gender and dependents columns were converted into binary numerical representations.

Variable	Original Values	Converted Values
Gender	Female/Male	0 / 1
Dependents	No/Yes	0 / 1

TABLE 1: CONVERTING CATEGORICAL VARIABLES INTO NUMERICAL VARIABLES

For Gender: Male = 1, Female = 0

For Dependents: Yes = 1, No = 0

This conversion allows the K-Means algorithm to process these variables numerically.

2.Handling Senior Citizen

The SeniorCitizen column had already been standardised in the processed dataset.

However, for clustering and easier business interpretation, the original binary values were restored from the raw dataset.

The values were maintained as:

- 0 = Not a senior citizen.
- 1 = Senior citizen.

This was done because the variable represents a simple yes/no characteristic, and retaining its original binary form makes the resulting cluster profiles easier to interpret.

3.Converting Boolean variables

The one-hot encoded columns were stored as Boolean values (True/False).

These columns were converted into integers:

- True = 1, False = 0

The transformed variables included:

- PhoneService_Yes
- MultipleLines_Yes
- InternetService_Fiber optic
- Contract_One year
- Contract_Two year

4.Handling the Churn variable

The Churn column was converted into binary values:

- Yes = 1, No = 0

However, an important distinction was maintained: churn was not included as an input feature in the clustering model.

Instead, it was retained separately to evaluate the churn rate of each customer segment after clustering.

These transformations ensured that the dataset was suitable for K-Means clustering while maintaining the original processed dataset for future predictive modelling. The transformations also allowed the resulting customer segments to be interpreted using understandable business characteristics.

4. Feature Selection and Scaling

Feature selection is an important stage because the variables included in a clustering model determine how customers are grouped.

The objective was to identify customers based on their characteristics and service usage rather than directly grouping them according to whether they had churned.

Excluding the Churn variable

The Churn variable was excluded from the clustering features because it represents the outcome that the business wants to understand. Including churn in the clustering process could cause the algorithm to separate customers based on whether they had already left or remained with the organisation.

This would reduce the usefulness of the analysis because the objective is to identify customer groups independently and then examine their churn behaviour.

Therefore, churn was kept aside and used later for segment-level comparison.

Excluding Gender

Gender was also excluded from the clustering model. Before excluding it, the churn rate was examined across both gender categories. The difference between the two groups was very small.

Gender	Churn Ratio
Female	26.9%
Male	26.2%

TABLE 2: GENDER CHURN RATIO

Therefore, gender was considered less useful for differentiating customers based on their churn behaviour and was excluded to avoid introducing unnecessary influence into the clustering process.

Selected clustering features

A total of nine features were selected for the final clustering model.

Feature	Description
Tenure	Length of time the customer has remained with the organisation
MonthlyCharges	Monthly amount paid by the customer
SeniorCitizen	Whether the customer is a senior citizen
Dependents	Whether the customer has dependents
PhoneService_Yes	Whether the customer has phone service
MultipleLines_Yes	Whether the customer has multiple phone lines
InternetService_Fiber optic	Whether the customer uses fibre internet
Contract_One year	Whether the customer has a one-year contract
Contract_Two year	Whether the customer has a two-year contract

Table 3: Selected clustering features

These variables were selected to capture a combination of customer tenure, spending behaviour, demographic characteristics and subscribed services.

Scaling the numerical variables

K-Means calculates distances between customers to determine which observations belong to similar groups. Variables measured on different scales can influence these distance calculations disproportionately.

For example, tenure is measured in months, whereas MonthlyCharges is measured in dollars.

However, the Data Engineer had already standardised these numerical variables during Part 1 of the data preparation process.

The descriptive statistics confirmed:

Variable	Mean	Standard Deviation
Tenure	0.00	1.00
Monthly Charges	0.00	1.00

TABLE 4: SCALING NUMERICAL VALUES

Therefore, no additional scaling was required for these two variables.

The remaining selected features were binary variables with values of 0 and 1, so they were retained in their existing form.

The feature selection process ensured that the model focused on customer characteristics that could help explain differences in customer behaviour.

Tenure and monthly charges were particularly important because they could help distinguish newly acquired customers from long-standing customers and lower-paying customers from premium-paying customers.

5. Determining the Optimal Number of Clusters

After selecting the features, the next step was to determine the appropriate number of customer segments.

K-Means requires the number of clusters (K) to be specified before training the model.

To identify a suitable value, two evaluation techniques were used:

1. Elbow Method.
2. Silhouette Score.

Both techniques were applied by testing different values of K, ranging from 1 to 10.

5.1 Elbow Method:

The Elbow Method evaluates how tightly customers are grouped around their respective cluster centres. It uses a measurement called inertia, also known as the within-cluster sum of squares. Lower inertia indicates that customers are positioned closer to their assigned cluster centres.

As the number of clusters increases, inertia generally decreases. However, adding too many clusters can make customer segmentation unnecessarily complicated.

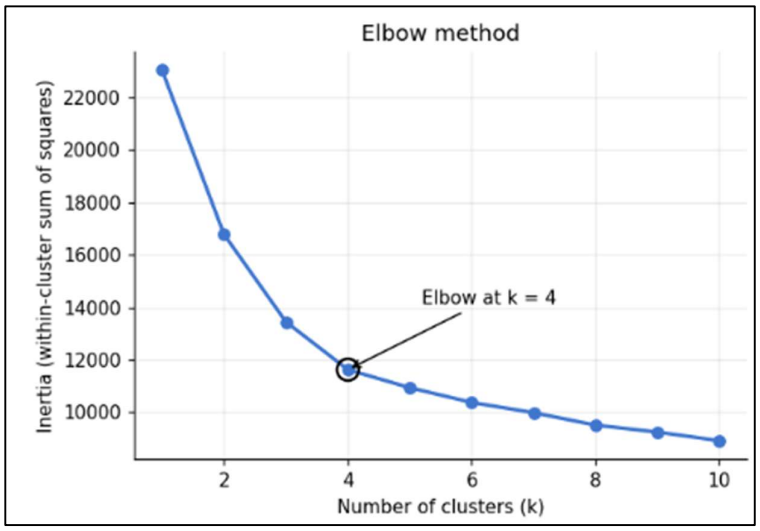


FIGURE 1: ELBOW METHOD

The objective is to identify the point where the reduction in inertia starts becoming smaller. The results were

Number of Clusters (K)	Inertia
1	23,041
2	16,777
3	13,442
4	11,620
5	10,930
6	10,368
7	9,978
8	9,504
9	9,238
10	8,910

TABLE 5: ELBOW METHOD

The inertia decreases significantly when moving from one cluster to four clusters.

After K = 4, the reduction becomes considerably smaller. This indicates that creating additional clusters beyond four provides less improvement in grouping customers.

Therefore, the elbow point was identified around K = 4.

5.2 Silhouette Score

The Silhouette Score was used as a second evaluation technique.

Number of Clusters (K)	Silhouette Score
2	0.246
3	0.232
4	0.223
5	0.200
6	0.178
7	0.161
8	0.162
9	0.153
10	0.163
2	0.246

TABLE 6: SILHOUETTE SCORE

It measures how closely each customer fits within their assigned cluster compared with the nearest alternative cluster.

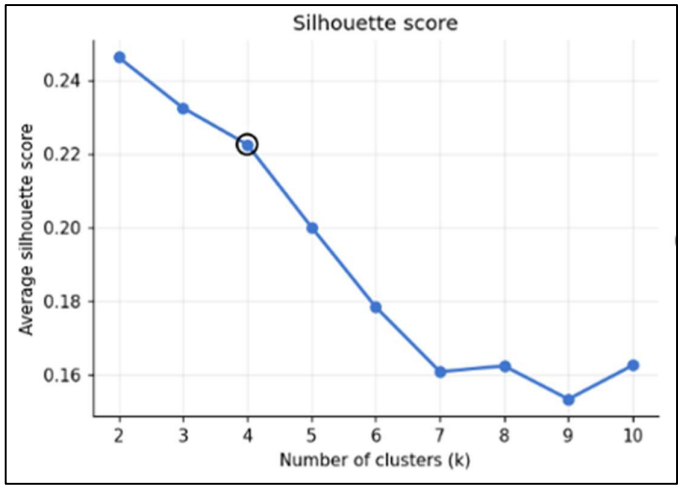


FIGURE 2: SILHOUETTE SCORE

The score ranges from -1 to +1:

- Close to +1: Customers are well matched to their assigned clusters.
- Around 0: Clusters overlap.
- Negative values: Some customers may be assigned to inappropriate clusters.

The highest silhouette score was obtained at K = 2, with a value of 0.246.

However, selecting two clusters would create a relatively broad segmentation and would not provide enough detail for developing different customer retention strategies.

At K = 4, the silhouette score was 0.223, which remained positive.

Considering both evaluation techniques, four clusters were selected as the final solution.

The decision was also supported by the business requirement to create a manageable number of customer segments that could be clearly interpreted and acted upon.

Although the silhouette score was not particularly high, this is understandable in customer segmentation, where customer characteristics can overlap rather than form perfectly separated groups.

Final selection: K = 4

Four clusters were selected because:

- The Elbow Method indicated a clear reduction in improvement beyond four clusters.
- The Silhouette Score remained positive.
- Four segments provided more meaningful business insights than two broad groups.
- Each segment demonstrated distinguishable customer characteristics.
- Four segments were manageable for designing targeted retention strategies.

6. Training the final K-Means Model

After determining the optimal number of clusters, the final K-Means model was trained using the nine selected features.

The following configuration was used:

Parameter	Value
Algorithm	K-Means
Number of clusters	4
Initialisations (n_init)	10
Random state	42
Input observations	7,043
Input features	9

TABLE 7: FINAL K-MEANS MODEL

The model was trained using the fit_predict() method, which identifies cluster centres and assigns each customer to a cluster. The final model produced the following evaluation results:

Evaluation Metric	Result
Inertia	11,619.6
Silhouette Score	0.225

TABLE 8: INERTIA AND SILHOUETTE SCORE

The inertia value represents the total within-cluster variation, while the silhouette score indicates the overall separation and cohesion of the resulting customer groups.

7. Model Stability Assessment

To check whether the clustering results were consistent, the model was retrained using different random states. The Adjusted Rand Index (ARI) was used to compare the resulting customer groupings.

ARI measures the similarity between two clustering solutions:

- 1.0 indicates identical groupings.
- 0 indicates similarity close to random assignment.
- Negative values indicate agreement below the expected random level.

The results were:

Random state	Adjusted Rand Index
0	1.000
1	0.999
7	0.999
21	0.999
99	0.999

TABLE 9: MODEL STABILITY ASSESSMENT

The ARI values were extremely close to 1.0.

This indicates that the clustering solution was highly consistent across different random initialisations.

It provides confidence that the four customer segments were not simply produced by an unusual random starting point.

8. Assigning Meaningful Cluster Numbers

K-Means initially assigns cluster labels automatically. These labels do not have any inherent business meaning. To make the results easier to interpret, the clusters were renumbered based on their average tenure and monthly charges. This produced a consistent order from Cluster 0 to Cluster 3.

The final customer distribution was:

Cluster	Number of customers
Cluster 0	1,721
Cluster 1	2,223
Cluster 2	1,176
Cluster 3	1,923
Total	**7,043**

TABLE 10: NUMBER OF CUSTOMERS IN EACH CLUSTER

9. Visualisation of Customer Clusters

Visualisation was used to understand the separation between customer groups and communicate the clustering results more effectively.

Two main visualisations were created.

9.1 Principal Component Analysis (PCA)

PCA was applied to reduce the nine-dimensional clustering dataset into two principal components. The purpose of PCA was to represent the customer segments in a two-dimensional graph.

PCA does not change the original clustering assignments. It transforms the data into a smaller number of dimensions for visual interpretation.

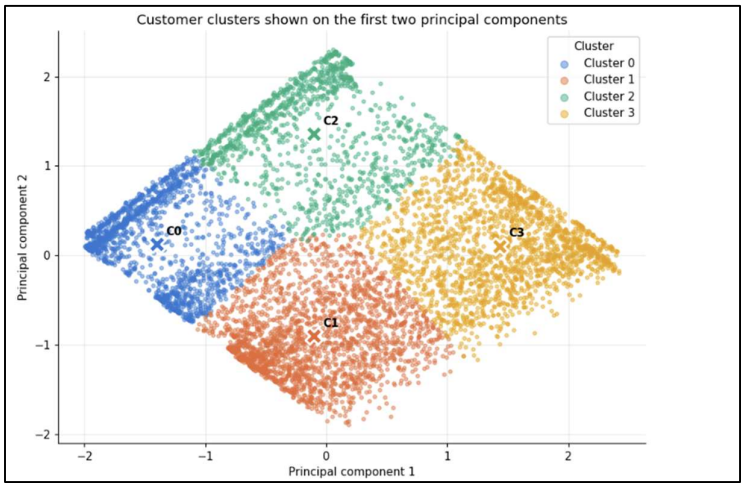


FIGURE 3: PRINCIPAL COMPONENT ANALYSIS (PCA)

The variance explained by the two principal components was:

Component	Variance explained
Principal Component 1	38.3%
Principal Component 2	23.7%

Combined	**62.0%**

TABLE 11: PRINCIPAL COMPONENT ANALYSIS (PCA)

The first two principal components explain 62% of the total variation in the original nine features. The visualisation displays individual customers using different colours according to their assigned clusters. The cluster centres are also highlighted.

The PCA visualisation demonstrates how the four customer segments are distributed across the reduced feature space. The clusters show distinguishable patterns, although some overlap is expected because customer behaviour is not perfectly separated.

It is important to recognise that the visualisation represents only 62% of the total variation, meaning that some information from the original nine dimensions is not displayed.

9.2 Clustering by Tenure and Monthly Charges

A second visualisation was created using the original customer data.It plots customer tenure against monthly charges, with different colours representing the four clusters.

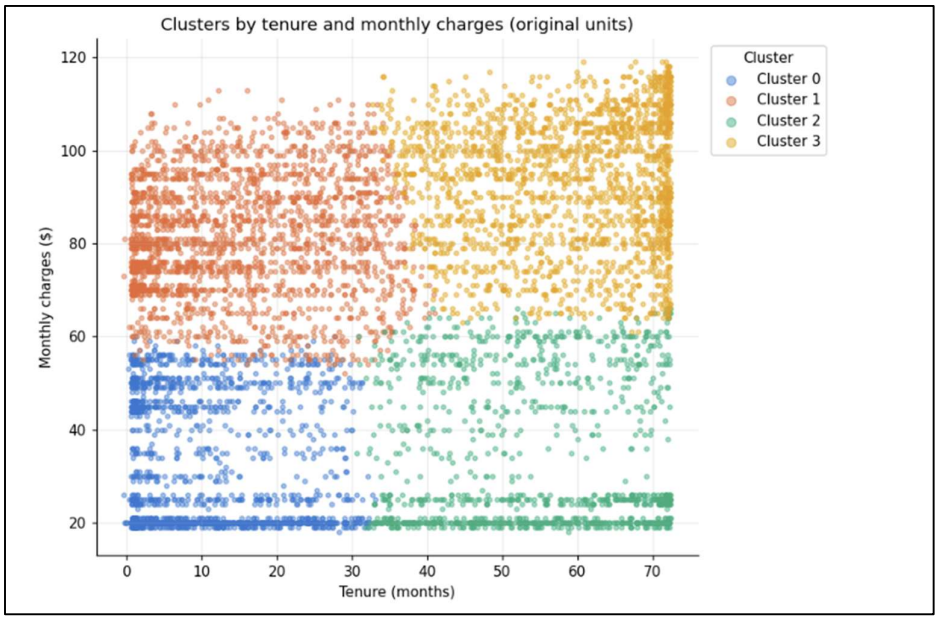


FIGURE 4: CLUSTERING BY TENURE AND MONTHLY CHARGES

This visualisation helps explain the actual business characteristics that distinguish the customer segments.

The graph reveals four broad combinations:

- Short tenure and low monthly charges.
- Short tenure and high monthly charges.
- Long tenure and low monthly charges.
- Long tenure and high monthly charges.

The visualisation indicates that tenure and monthly charges are the main factors distinguishing the four customer groups. Newer customers are generally separated from long-standing customers, while monthly

charges distinguish budget customers from premium customers. This provides a clearer business interpretation of the clustering results.

10. Customer Segment Profiling

After assigning customers to clusters, the next step was to understand the characteristics of each segment. The raw dataset was combined with the assigned cluster labels to calculate customer profiles.

The profiling process examined:

- Number of customers.
- Percentage of total customers.
- Average tenure.
- Average monthly charges.
- Churn rate.
- Percentage of senior citizens.
- Percentage of customers with dependents.
- Phone service subscriptions.
- Multiple-line subscriptions.
- Fibre internet subscriptions.
- Contract distribution.

This stage is particularly important from a business analyst's perspective because clustering alone only identifies groups. Profiling explains what those groups represent and how the business can respond.

11. Overall Customer Segment Distribution

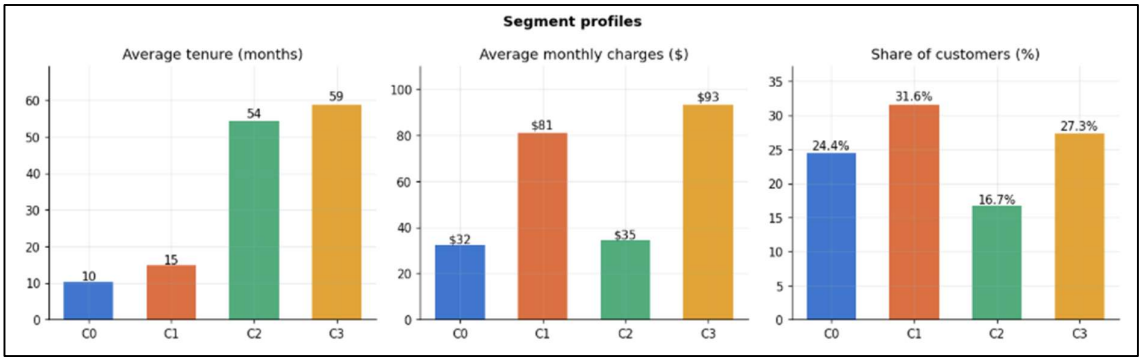


FIGURE 5: SEGMENT PROFILES

Cluster	Customer segment	Customers	Share
C0	New Budget Customers	1,721	24.4%
C1	At-Risk New Premium Customers	2,223	31.6%

C2	Loyal Budget Customers	1,176	16.7%
C3	Loyal Premium Customers	1,923	27.3%

TABLE 12: CUSTOMER SEGMENTS

The largest segment is Cluster 1, representing 31.6% of the total customer base. Cluster 2 is the smallest segment, representing 16.7%.

11.1 Cluster 0 – New Budget Customers

Customer profile:

- Number of customers: 1,721.
- Customer share: 24.4%.
- Average tenure: 10.4 months.
- Average monthly charges: $32.40.
- Churn rate: 24.6%.

This segment represents customers who have recently joined the organisation and generally pay lower monthly charges. Their average tenure of approximately 10 months suggests that they are still relatively new customers. Approximately 54.6% are on month-to-month contracts, while 44.3% use fibre internet.

The churn rate is 24.6%, which is slightly below the overall customer churn rate. These customers represent an important opportunity for improving customer retention during the early stages of their relationship with the organisation. Although they generally pay lower monthly charges, they have not yet developed the same level of loyalty as long-standing customers.

Recommended business actions:

- Introduce onboarding programmes for newly acquired customers.
- Provide incentives for customers to continue their subscriptions.
- Monitor customer satisfaction during the first year.
- Offer suitable service packages based on customer requirements.

11.2 Cluster 1 – At-Risk New Premium Customers

Customer profile:

- Number of customers: 2,223.
- Customer share: 31.6%.
- Average tenure: 14.9 months.
- Average monthly charges: $81.20.
- Churn rate: 49.0%.

This segment represents the highest-risk customer group identified in the analysis. These customers have relatively short tenure but pay significantly higher monthly charges than the other segments.

The churn rate is 49.0%, meaning almost half of the customers in this segment have churned. Approximately 55.6% are on month-to-month contracts. This segment also has a higher proportion of senior citizens (23.8%) compared with Cluster 0 and Cluster 2.

From a business perspective, this is the most important segment for retention activities. Customers are paying premium prices but have not developed strong long-term relationships with the organisation. High monthly charges combined with relatively short tenure may indicate that some customers are not receiving sufficient perceived value from their subscriptions. However, the clustering results identify an association rather than proving that high charges directly cause churn.

Recommended business actions:

- Prioritise this segment for targeted retention campaigns.
- Review pricing and service-package value.
- Introduce personalised discounts or loyalty incentives.
- Contact customers proactively to identify dissatisfaction.
- Provide alternative packages that better match customer requirements.
- Monitor this segment regularly using customer churn indicators.

This group should be the first priority when developing customer retention strategies.

11.3 Cluster 2 – Loyal Budget Customers

Customer profile:

- Number of customers: 1,176.
- Customer share: 16.7%.
- Average tenure: 54.2 months.
- Average monthly charges: $34.50.
- Churn rate: 4.8%.

This segment represents long-standing customers who generally pay lower monthly charges.

Their average tenure is approximately 54 months, demonstrating a relatively established relationship with the organisation. This segment has the lowest churn rate across all four clusters.

Approximately 45.4% of these customers have dependents, making it the segment with the highest proportion of customers with dependents. These customers represent a stable customer group with strong retention performance. Their relatively low monthly charges and long tenure suggest that customers can remain loyal without necessarily subscribing to premium-priced services. This segment provides an opportunity to understand which service characteristics and customer experiences support long-term retention.

Recommended business actions:

- Maintain competitive and affordable pricing.
- Introduce loyalty rewards to recognise long-term customers.

- Avoid unnecessary price increases.
- Offer additional services where they provide genuine customer value.
- Use this segment as a reference when examining long-term customer retention.

11.4 Cluster 3 – Loyal Premium Customers

Customer profile:

- Number of customers: 1,923.
- Customer share: 27.3%.
- Average tenure: 58.8 months.
- Average monthly charges: $93.30.
- Churn rate: 15.5%.

This segment represents customers with the highest average tenure and the highest monthly charges. They have maintained their subscriptions for approximately 59 months and contribute a relatively high monthly revenue per customer. Almost all customers in this segment have phone service.

Approximately 45.5% use fibre internet, while 55.3% have month-to-month contracts. Despite paying the highest monthly charges, this segment has a relatively low churn rate of 15.5%.

Business interpretation:

These customers represent a valuable segment because they combine long-term relationships with higher monthly spending.

Their relatively low churn rate suggests that customers who remain with the organisation over a longer period may develop stronger loyalty.

Retaining these customers is important because losing them could result in a greater loss of monthly revenue compared with losing lower-paying customers.

12. Recommended business actions:

- Introduce premium loyalty programmes.
- Provide personalised customer service.
- Offer additional services and upgrades where appropriate.
- Monitor satisfaction to maintain their loyalty.
- Develop targeted retention activities for high-value customers.

13. Churn Analysis Across Customer Segments

One of the most important objectives of the clustering analysis was to understand how churn differs across the four customer groups.

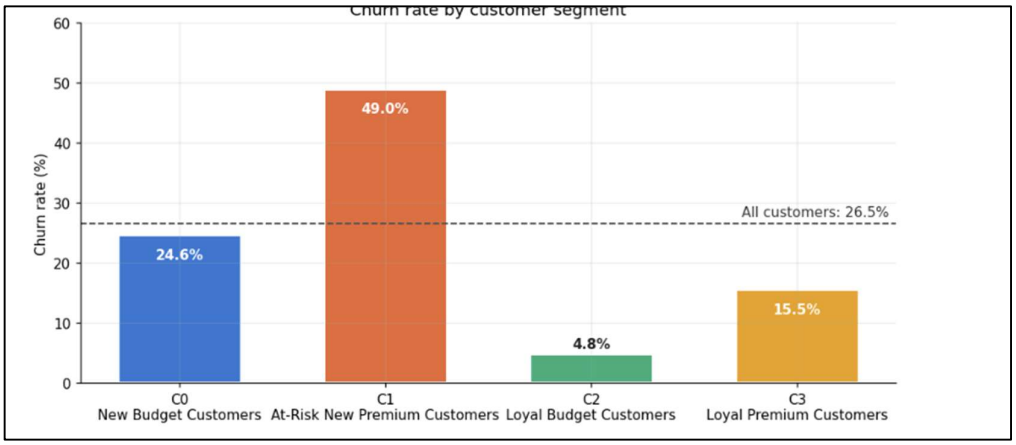


FIGURE 6: CHURN RATE BY CUSTOMER SEGMENT

After assigning customers to clusters, the churn rate was calculated separately for each segment.

The results were:

Cluster	Segment	Churn rate
C0	New Budget Customers	24.6%
C1	At-Risk New Premium Customers	49.0%
C2	Loyal Budget Customers	4.8%
C3	Loyal Premium Customers	15.5%

TABLE 13: CHURN RATE FOR EACH CLUSTER

The overall churn rate across the dataset was approximately 26.5%.

KEY FINDINGS

1. New customers demonstrate higher churn risk.

Clusters 0 and 1 consist of customers with relatively short tenure.

These two groups demonstrate higher churn rates compared with the long-standing customer segments.

This suggests that the early customer lifecycle is an important period for retention activities.

2. New premium customers are the highest-risk segment.

Cluster 1 has a churn rate of 49.0%, which is substantially higher than the overall churn rate.

This indicates that customers paying higher monthly charges during the early stages of their relationship may require additional attention.

3. Long-standing customers demonstrate stronger retention.

Clusters 2 and 3 have significantly lower churn rates.

Cluster 2 has the lowest churn rate at only 4.8%, while Cluster 3 has a churn rate of 15.5%.

This suggests that long-term customer relationships are associated with better retention outcomes.

4. Premium customers represent an important revenue opportunity.

Cluster 3 has the highest average monthly charges at $93.30.

Although its churn rate is relatively low, retaining these customers remains important because of their higher monthly spending.

5. Contract type and internet service do not strongly differentiate the segments.

The percentage of month-to-month contracts is relatively consistent across all four clusters, ranging from 54.2% to 55.6%.

Similarly, fibre internet subscriptions range from approximately 40.3% to 45.5%.

This suggests that tenure and monthly charges provide more useful distinctions between the customer groups than these particular service and contract variables in the resulting profiles.

10. Overall Business Insights and Recommendations

From the clustering analysis, several important business insights can be identified.

10.1 Prioritise early-stage customer retention

The two newer customer segments account for a significant proportion of the customer base and demonstrate higher churn rates.

The organisation should focus on improving customer onboarding, engagement and satisfaction during the initial customer lifecycle.

10.2 Focus on high-risk premium customers

Cluster 1 should be the highest priority for customer retention initiatives.

With a churn rate of 49.0%, this group represents a substantial opportunity to reduce customer losses.

The organisation should review pricing, service value and customer satisfaction within this segment.

10.3 Maintain relationships with loyal customers

Clusters 2 and 3 demonstrate relatively strong retention.

The organisation should continue providing consistent service quality and introduce loyalty initiatives to protect these customer relationships.

10.4 Protect high-value customers

Cluster 3 represents an important revenue-generating customer group because of its high average monthly charges.

Even though the churn rate is relatively low, targeted engagement can help protect recurring revenue.

10.5 Develop segment-specific marketing strategies

Instead of applying a single marketing strategy to the entire customer base, the organisation can use the four customer segments to develop more targeted campaigns.

This can improve the relevance of retention offers and help allocate marketing resources more effectively.

11. Exporting and Saving the Clustering Results

After completing the analysis, the clustering results were exported into separate CSV files to support further business analysis and predictive modelling.

Three main datasets were generated.

11.1 Clustered_Customers.csv

This file contains the original customer attributes along with the assigned cluster number and customer segment name.

It is primarily intended for the Business Analyst to examine individual customer segments and develop retention strategies.

11.2 Updated_Dataset_with_Clusters.csv

This file contains the processed dataset with an additional Cluster column.

It was prepared for the Data Analyst responsible for predictive modelling.

The cluster information can be tested as an additional input feature in the Artificial Neural Network model.

11.3 Cluster_Profiles.csv

This file contains the summary statistics for each customer segment.

It includes the customer count, average tenure, monthly charges, churn rate and other segment characteristics.

It can be used for reporting, visualisation and business interpretation.

The analysis also generated five visualisations covering the elbow and silhouette methods, PCA clusters, tenure versus monthly charges, churn by cluster and segment profiles.

12. Saving and Validating the K-Means Model

The final trained K-Means model was saved using the Joblib library.

The model was saved as:

kmeans_k4_model.joblib

Along with the trained model, the following information was stored:

- Trained K-Means model.
- Selected clustering features.
- Cluster relabelling information.
- Customer segment names.

This allows the team to reload the trained model and assign cluster labels to new customer records without having to retrain the entire model. A validation check was also performed by reloading the saved model and comparing its predictions with the original cluster assignments.

The validation result was **Labels reproduced: True**

This confirms that the saved model successfully reproduced the original customer cluster assignments.

Limitations of the Analysis

Although the clustering analysis provides useful customer segmentation insights, some limitations should be considered.

1. Limited dataset attributes

The dataset contains only ten original attributes. Additional information such as customer usage patterns, billing history, payment methods and customer service interactions could provide further insights.

2. Overlapping customer characteristics

Customer behaviour is complex, and individuals do not always fall into clearly separated groups.

The silhouette score of 0.225 indicates that some overlap exists between the customer segments.

Therefore, the clusters should be interpreted as broad customer groups rather than perfectly separated categories.

3. K-Means assumptions

K-Means works best when clusters have relatively compact and similarly shaped distributions.

Real customer behaviour may not always follow these patterns, which can affect the clustering results.

4. Single dataset snapshot

The analysis is based on a single snapshot of customer information.

It does not show how customer behaviour changes over time or how customers move between segments.

5. Correlation does not establish causation

Although some segments demonstrate higher churn rates, the clustering analysis does not establish that a particular feature directly causes customers to leave.

Conclusion

The clustering analysis successfully identified four distinct customer segments using the K-Means algorithm.

The model was developed using 7,043 customer records and nine selected features covering tenure, monthly charges, demographic characteristics and subscribed services.

The Elbow Method and Silhouette Score were used to evaluate the appropriate number of clusters, resulting in the selection of four customer segments.

The final analysis identified the following groups:

1. New Budget Customers.
2. At-Risk New Premium Customers.
3. Loyal Budget Customers.

4. Loyal Premium Customers.

The most significant finding was that the At-Risk New Premium Customers segment has the highest churn rate at 49.0%.

In comparison, Loyal Budget Customers demonstrate the strongest retention performance, with a churn rate of only 4.8%.

These findings highlight the importance of customer tenure and monthly charges in understanding different customer behaviours.

From a business analyst's perspective, the results provide a foundation for developing targeted retention strategies, improving customer engagement and protecting recurring revenue.

The clustering outputs have also been prepared for the next stage of the project, where customer segmentation information can be evaluated alongside predictive modelling.

Overall, the clustering analysis transforms customer-level data into meaningful business segments, allowing the organisation to make more informed and targeted customer retention decisions.

NEXT STEPS

Following the completion of the clustering analysis, the next stage of the project will focus on predictive modelling and interpreting the results from the Artificial Neural Network (ANN).

The cluster information will be shared with the Data Analyst to investigate whether customer segments can improve predictive performance.

The next stage will include:

- Completing the ANN predictive modelling implementation.
- Evaluating the model's ability to identify customers likely to churn.
- Interpreting the predictive modelling results.
- Comparing the predictive findings with the clustering results.
- Answering the key business questions established for the project.
- Developing final recommendations based on both descriptive and predictive analysis.

Combining clustering and predictive modelling will provide a more comprehensive understanding of customer churn and support evidence-based business decisions.